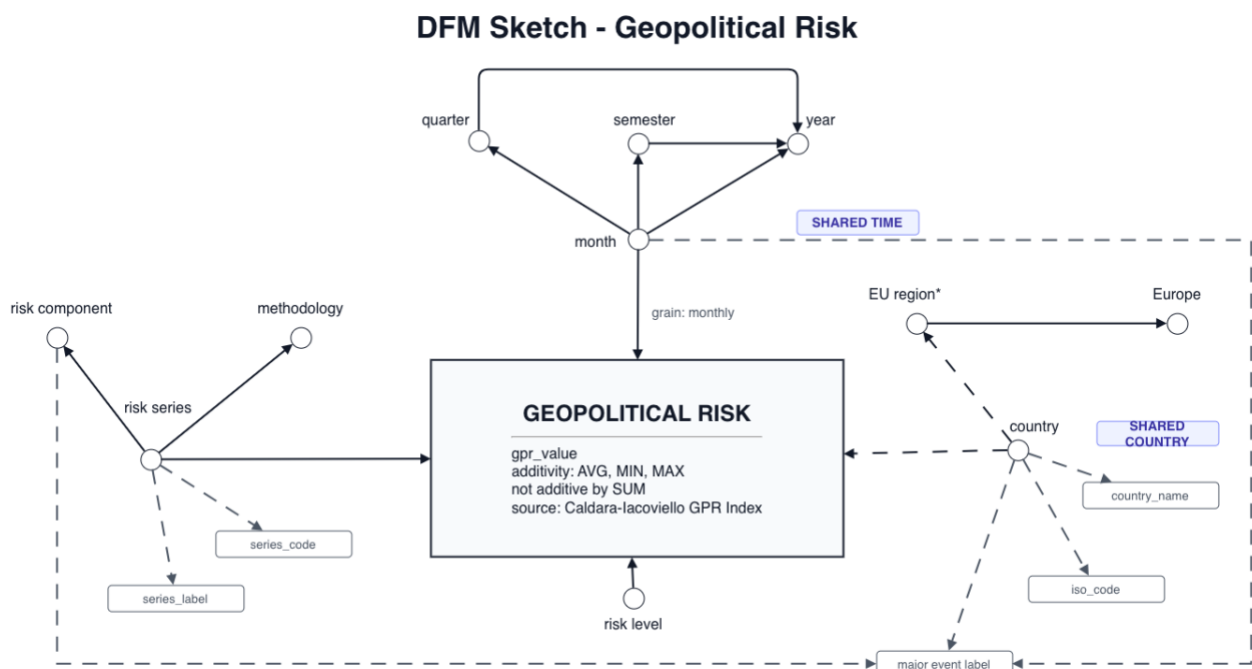


DFM SKETCH

The following DFM sketches describe the preliminary multidimensional design of the energy-related data warehouse. Each schema is centered on a fact representing a specific analytical topic and includes the main measures, dimensions, hierarchies, descriptive attributes, and shared dimensions. Time and country are modeled as common dimensions in order to support integrated analysis and drill-across queries across the different facts.

DFM Sketch – Geopolitical Risk

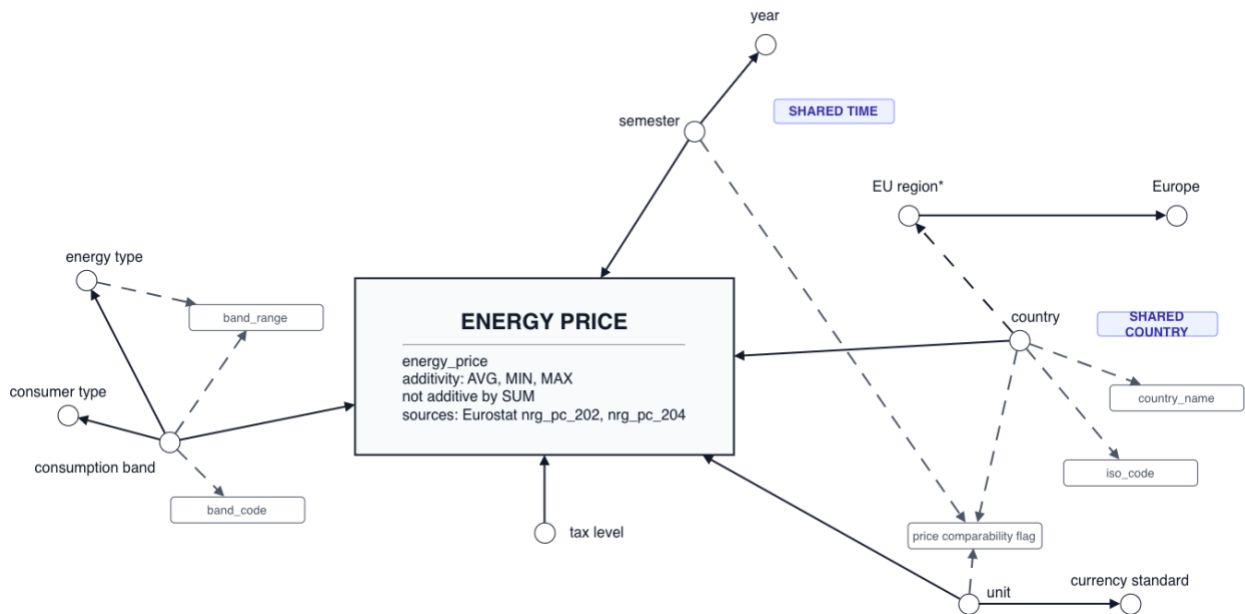
This DFM models the geopolitical risk indicator over time, using **gpr_value** as the central measure. The main dimensions are time, risk series, and country, the latter being optional since global GPR series are not associated with a specific country. Risk level is a derived dimension, generated during the ETL process from **gpr_value** thresholds. Since the measure is not additive by sum, it should be analyzed using aggregation functions such as average, minimum, and maximum.



DFM Sketch – Energy Price

This DFM represents energy prices by country, semester, consumption band, tax level, and unit. Energy type and consumer type are attributes determined by consumption band rather than independent dimensions. The central measure is **energy_price**, which is non-additive by sum and should be analyzed using average, minimum, and maximum values. Price comparability flags and currency standard attributes support correct interpretation and comparison of prices across countries and periods.

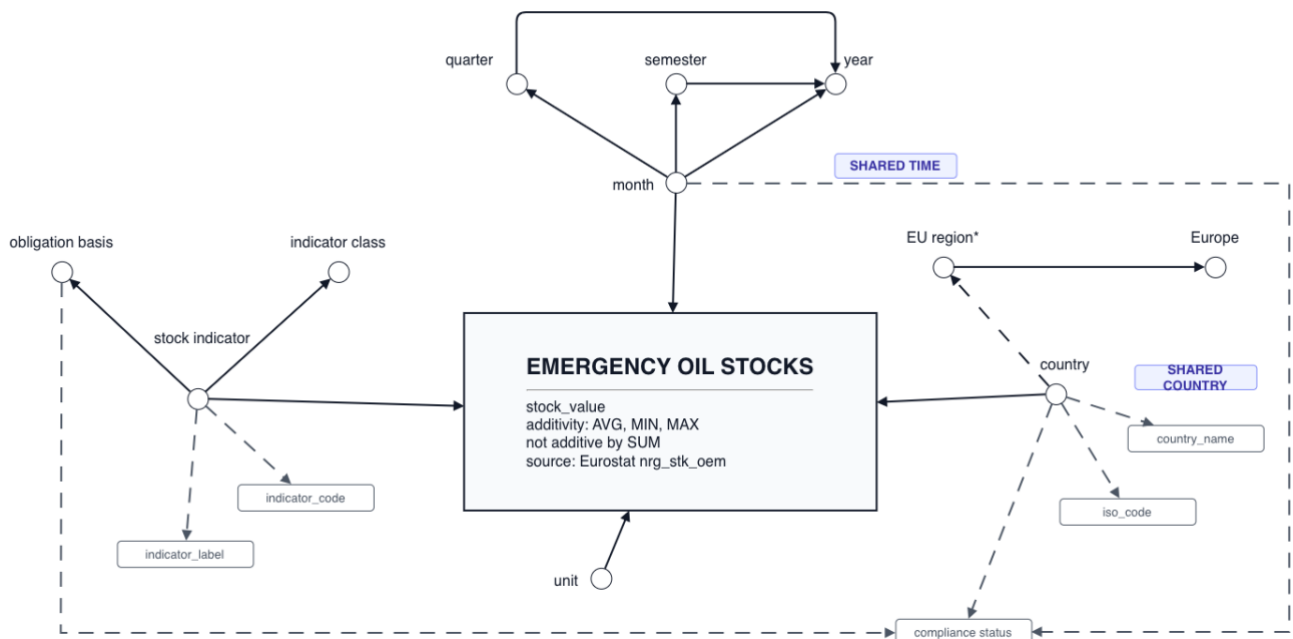
DFM Sketch - Energy Price



DFM Sketch - Emergency Oil Stocks

This DFM describes emergency oil stock values reported by European countries over time. The main dimensions are time, country, stock indicator, and unit; obligation basis is modeled as a hierarchy level of stock indicator, while compliance status is a cross-dimensional attribute depending on country, month, and obligation basis. The central measure is **stock_value**, which is not additive by sum and can be analyzed using average, minimum, and maximum values.

DFM Sketch - Emergency Oil Stocks



DFM Sketch - Import Dependency

This DFM models the import dependency rate of energy products for European countries on an annual basis. The central fact contains the measure **import_dependency_rate**, which is non-additive by sum and can be analyzed using average, minimum, and maximum values. The main dimensions are time, country, and energy product, with a product hierarchy from energy product to product category and energy family. The model also includes country-related attributes and EU membership information to support geographical and institutional analysis.

DFM Sketch - Import Dependency

